Transportation Method APP

QUARTER	EXPECTED DEMAND	REGULAR CAPACITY	OVERTIME CAPACITY	SUBCONTRACT CAPACITY
1	900	1000	100	500
2	1500	1200	150	500
3	1600	1300	200	500
4	3000	1300	200	500

Regular production cost per unit $20
Overtime production cost per unit $25
Subcontracting cost per unit $28
Inventory holding cost per unit per period $3
Beginning inventory 300 units

1

Transportation Tableau

	PERIOD OF PRODUCTION	PERIOD OF USE 1	2	3	4	Unused Capacity	Capacity
	Beginning Inventory	0	— 3	— 6	— 9		300
1	Regular	20	23	26	— 29		1000
	Overtime	25	28	31	34		100
	Subcontract	28	31	34	37		500
2	Regular		20	— 23	— 26		1200
	Overtime		25	28	31		150
	Subcontract		28	31	34		500
3	Regular			20	— 23		1300
	Overtime			25	— 28		200
	Subcontract			28	31		500
4	Regular				20		1300
	Overtime				25		200
	Subcontract				28		500
	Demand	900	1500	1600	3000		

2

Transportation Tableau

	PERIOD OF PRODUCTION	PERIOD OF USE 1	2	3	4	Unused Capacity	Capacity
	Beginning Inventory	300 0	— 3	— 6	— 9		300
1	Regular	600 20	300 23	100 26	— 29		1000
	Overtime	25	28	31	100 34		100
	Subcontract	28	31	34	37		500
2	Regular		1200 20	— 23	— 26		1200
	Overtime		25	28	150 31		150
	Subcontract		28	31	250 34	250	500
3	Regular			1300 20	— 23		1300
	Overtime			200 25	— 28		200
	Subcontract			28	500 31		500
4	Regular				1300 20		1300
	Overtime				200 25		200
	Subcontract				500 28		500
	Demand	900	1500	1600	3000		

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Production Plan

PERIOD	DEMAND	REGULAR PRODUCTION	OVERTIME	SUB-CONTRACT	ENDING INVENTORY
1	900	1000	100	0	500
2	1500	1200	150	250	600
3	1600	1300	200	500	1000
4	3000	1300	200	500	0
Total	7000	4800	650	1250	2100

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Master Production Schedule (MPS)

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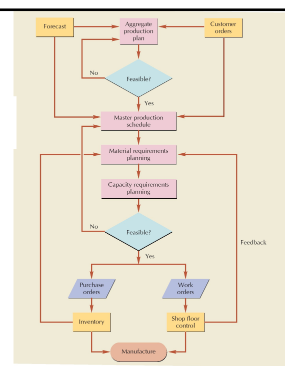
Resource Planning for Manufacturing



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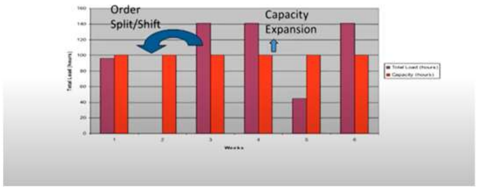
Checking Capacity Feasibility

For each workcenter …



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Aggregate Production Plan

Months	January				February			
Aggregate Production Plan (Shows the total quantity of amplifiers)	1,500				1,200			
Weeks	1	2	3	4	5	6	7	8
Master Production Schedule (Shows the specific type and quantity of amplifier to be produced)								
240-watt amplifier	100		100		100		100	
150-watt amplifier		500		500		450		450
75-watt amplifier			300				100	

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Master Production Schedule

- MPS sets the quantity of each end item to be completed in each week of the short range planning horizon.
- End item are finished product, or parts that are shipped as end item. End item may be shipped to customer or placed in inventory.
- The MPS translates the business plan, including forecast demand, into a production plan using planned orders in a true multi-level optional component scheduling environment.
- MPS helps in avoiding shortages, costly expediting, last minute scheduling, and inefficient allocation of resources.

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Objectives of MPS

- Determine the quantity and timing of completion of end items over a short-range planning horizon.
- Schedule end items (finished goods and parts shipped as end items) to be completed promptly and when promised to the customer.
- Avoid overloading or underloading the production facility so that production capacity is efficiently utilized and low production costs result.

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Master Production Schedule (MPS)

- ☑ MPS is established in terms of specific products
- ☑ Schedule must be followed for a reasonable length of time
- ☑ The MPS is quite often fixed or frozen in the near term part of the plan
- ☑ The MPS is a rolling schedule
- ☑ The MPS is a statement of what is to be produced, not a forecast of demand

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Master Production Schedule (MPS)

- ☑ Specifies what is to be made and when
- ☑ Must be in accordance with the aggregate production plan
- ☑ Inputs from financial plans, customer demand, engineering, supplier performance
- ☑ As the process moves from planning to execution, each step must be tested for feasibility
- ☑ The MPS is the result of the production planning process

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Material Management
Material Requirement Planning

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Materials Requirement Planning

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What is MRP?

MRP is the approach of determining the number of parts, and materials required to produce each end item. MRP provides the schedule specifying when each of these items should be ordered or produced.

01 Universally adopted in manufacturing firms

02 Based on dependent demand

03 Straight forward, simple, logical process to estimate demand for dependent parts

04 Used in industries where products are made in batches using the same equipment

05 Not useful for producing complex, expensive, and advanced research products

06 Generally implemented to meet the short-term aggregate production plan.

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Materials Requirement Planning

Dependent Demands

When demand for items is derived from plans to make certain products, as it is with raw materials, parts, and assemblies used in producing a finished product, those items are said to have dependent demand.

The parts and materials that go into the production of cars are examples of dependent demand because the total quantity of parts and raw materials needed during any time period depends on the number of cars that will be produced. Conversely, demand for the finished cars is independent—a car is not a component of another item.

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Material Requirements Planning (MRP)

A methodology used for planning the production of assembled products such as smartphones, automobiles, kitchen tables, and a whole host of other products that are assembled.

- Some items are produced repetitively while others are produced in batches.

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What is MRP?

- Computerized Inventory Control
- Production Planning System
- Management Information System
- Manufacturing Control System

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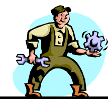
When to use MRP

- Job Shop Production
- Complex Products
- Assemble-to-Order Environments
- Discrete and Dependent Demand Items



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Where MRP can be used

Most valuable in industries where the products are made in batches using the same productive equipment

Industry Type	Examples	Expected Benefits
Assemble-to-stock	Combines multiple component parts into a finished product, which is then stocked in inventory to satisfy customer demand. Examples: Watches, tools, appliances	High
Make-to-stock	Items are manufactured by machine rather than assembled from parts. These are standard stock items carried in anticipation of customer demand. Examples: piston rings, electrical switches.	Medium
Assemble-to-order	A final assembly is made from standard options that the customer chooses. Examples: Trucks, generators, motors.	High
Make-to-order	Items are manufactured by machine to customer order. These are generally industrial orders. Examples: bearings, gears, fasteners.	Low
Engineer-to-order	Items are fabricated or assembled completely to customer specification. Examples: Turbine generators, heavy machine tools.	High
Process	Includes industries such as foundries, rubber and plastics, specialty paper, chemicals, paint, drug, food processors.	Medium

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What can MRP do?

- Reduce Inventory Levels
- Reduce Component Shortages
- Improve Shipping Performance
- Improve Customer Service
- Improve Productivity
- Simplified and Accurate Scheduling
- Reduce Purchasing Cost
- Improve Production Schedules
- Reduce Manufacturing Cost
- Reduce Lead Times
- Less Scrap and Rework
- Higher Production Quality

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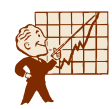
What can MRP do?

- Improve Communication
- Improve Plant Efficiency
- Reduce Freight Cost
- Reduction in Excess Inventory
- Reduce Overtime
- Improve Supply Schedules
- Improve Calculation of Material Requirements
- Improve Competitive Position



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Three Basic Steps of MRP

Steps

- Identifying Requirements
- Running MRP – Creating the Suggestions
- Firming the Suggestions

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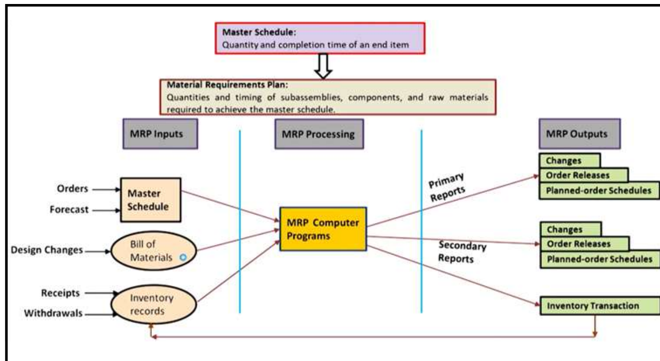

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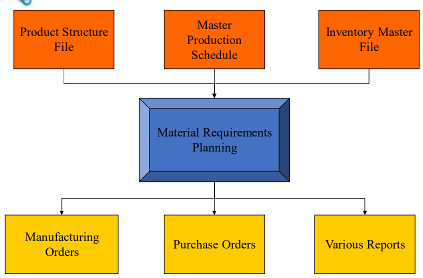
Overview of the MRP System



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Step 1: Identifying the Requirements

- Quantity on Hand
- Quantity on Open Purchase Order
- Quantity in/or Planned for Manufacturing
- Quantity Committed to Existing Orders
- Quantity Forecasted

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- Critical Items
- Expedite Items
- Delay Items

Step 2: Running MRP – Creating the Suggestions

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Step 3: Firming the Suggestions

- Manufacturing Orders
- Purchasing Orders
- Various Reports

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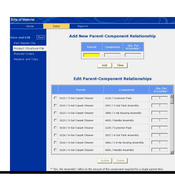
Product Structure File

- The Product Structure File describes each parent-component relationship, using the following information:
  - Part number for the parent item
  - Part number for the component
  - Quantity of the component used in the parent item
- Bill of Materials



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Master Production Schedule

- Schedule of Finished Products
- Represents Production, not Demand
- Combination of Customer Orders and Demand Forecasts
- What Needs to be Produced

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MPS provides basic foundation for

- Planning for the material and capacity requirements
- Making good use of manufacturing resources
- Making customer delivery promises
- Resolving tradeoffs between sales and manufacturing and
- Attaining strategic objectives in the sales and operations plan

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Material Requirements Planning System Structure

Demand for products

Sources:
- Customer orders (no forecasting involved)
- Aggregate production plan (Implemented through detailed master production schedule)

Bill-of-Materials
Complete product description

Three main inputs of MRP Program
- Bill-of-Materials (BOM) file
- Master schedule file
- Inventory records file

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BOM is a comprehensive inventory of the raw materials, assemblies, subassemblies, parts and components, as well as the quantities of each, needed to manufacture a product

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Types of BOM

There are several functional types of BOM

ENGINEERING BILL OF MATERIALS
It is used for the creation of a new finished good. It is the ground zero for a finished product that lists all parts, components, and materials for the finished product as it was originally designed.

MANUFACTURING BILL OF MATERIALS
It is the most recognized form and consists of all materials, assemblies, formulas, or components required to produce a shippable product.

CONFIGURABLE BILL OF MATERIALS
many manufacturing companies produce the same product in a variety of sizes, colors, or other parameters. Core product may be same, but the final version may differ slightly depending on the customer

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Inventory Master File

- On-Hand Quantities
- On-Order Quantities
- Lot Sizes
- Safety Stock
- Lead Time
- Past-Usage Figures

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MRP INPUTS

The Master Schedule

- Which end items are to be produced, when they are needed, and in what quantities.

Week Number

Item:X	1	2	3	4	5	6	7	8
Quantity				100				150

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The Bill of Materials

A bill of materials (BOM) contains a listing of all of the assemblies, subassemblies, parts, part costs, and raw materials that are needed to produce one unit of a finished product. Thus, each finished product has its own bill of materials

The listing in the bill of materials is hierarchical; it shows the quantity of each item needed to complete one unit of its parent item. The nature of this aspect of a bill of materials is clear when you consider a product structure tree, which provides a visual depiction of the subassemblies and components needed to assemble a product.

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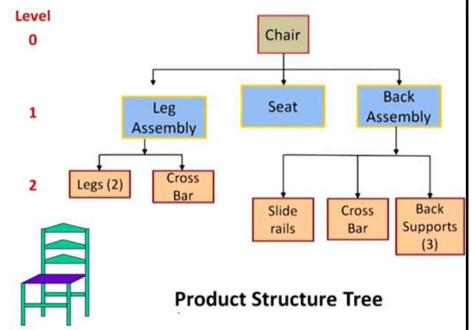
Bill of Materials

Level		Quantity
0	Chair	1
1	Leg assembly	1
2	Legs	2
2	Cross bar	1
1	Seat	1
1	Back assembly	1
2	Slide rails	2
2	Cross bar	1
2	Back supports	3



Product Structure Tree

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Example Problem

Determining How much of Each Component will be needed for Assembly-

Use the information presented in Figure to do the following:
a. Determine the quantities of B, C, D, E, and F needed to assemble one X.
b. Determine the quantities of these components that will be required to assemble 10 Xs, taking into account the quantities on hand (i.e., in inventory) of various components:

B	4
C	10
D	8
C	60

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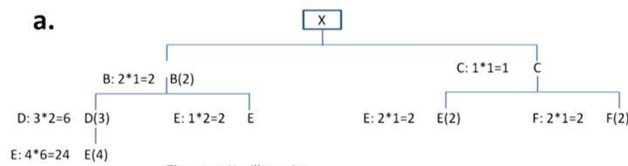
Problem Solution

a.



Thus, one X will require
B: 2
C: 1
D: 6
E: 28(Note that E occurs in three places, with requirements of 24 + 2 + 2 = 28)
F: 2

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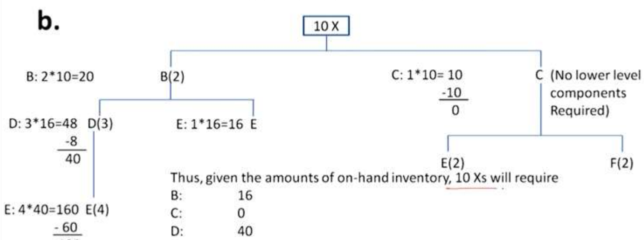
Problem Solution

b.



Thus, given the amounts of on-hand inventory, 10 Xs will require
B: 16
C: 0
D: 40
E: 116
F: 0

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BENEFITS AND REQUIREMENTS OF MRP-

BENEFITS:-

1. Low levels of in-process inventories, due to an exact matching of supply to demand.
2. The ability to keep track of material requirements.
3. The ability to evaluate capacity requirements generated by a given master schedule.
4. A means of allocating production time.
5. The ability to easily determine inventory usage by **backflushing.**

REQUIREMENTS:-

1. A computer and the necessary software programs to handle computations and maintain records
2. Accurate and up-to-date:
a. Master schedules
b. Bills of materials
c. Inventory records
3. Integrity of file data

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MRP PROCESSING

MRP processing takes the end item requirements specified by the master schedule and "explodes" them into time-phased requirements for assemblies, parts, and raw materials using the bill of materials offset by lead times.

- MRP processing combines the time phasing and "explosion" into a sequence of spreadsheet sections.

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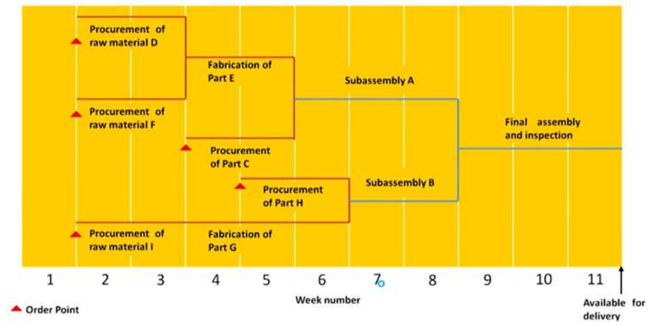

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Terms Used in MRP Processing

- **Gross requirements**
- **Scheduled receipts**
- **Projected on hand**
- **Net requirements**
- **Planned-order receipts**
- **Planned-order releases**

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Example Problem (1)

MRP schedule with lot-for-lot ordering-

Master schedule for shutters:

Week number	Beg. Inv.	1	2	3	4	5	6	7	8
Quantity					100				150

Item		Beg. Inv.	1	2	3	4	5	6	7	8
Shutters: LT = 1 week	Gross requirements					100				150
	Scheduled receipts									
	Projected on hand									
	Net requirements					100				150
	Planned-order receipts					100				150
	Planned-order releases			times	100			times	150	
Frames: LT = 2 weeks	Gross requirements			2	200			2	300	
	Scheduled receipts									
	Projected on hand									
	Net requirements				200				300	
	Planned-order receipts				200				300	
	Planned-order releases		200	times			300	times		
Wood sections: LT = 1 week	Gross requirements			4	400			4	600	
	Scheduled receipts		70							
	Projected on hand		70	70	70					
	Net requirements				330				600	
	Planned-order receipts				330				600	
	Planned-order releases			330				600		

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Example Solution (Explanation)

- Therefore, planned receipts for week 4 equal 100 shutters. Because shutter assembly requires one week, this means a planned-order release at the start of week 3.
- Using the same logic, 150 shutters must be assembled during week 7 in order to be available for delivery at the start of week 8.
- The planned-order release of 100 shutters at the start of week 3 means that 200 frames (gross requirements) must be available at that time. Because none are expected to be on hand, this generates net requirements of 200 frames and necessitates planned receipts of 200 frames by the start of week 3.
- With a two-week lead time, this means that the firm must order 200 frames at the start of week 1. Similarly, the planned-order release of 150 shutters at week 7 generates gross and net requirements of 300 frames for week 7 as well as planned receipts for that time.
- The two-week lead time means the firm must order frames at the start of week 5.
- The planned-order release of 100 shutters at the start of week 3 also generates gross requirements of 400 wood sections at that time.
- However, because 70 wood sections are expected to be on hand, net requirements are 400 — 70 = 330. This means a planned receipt.

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Example Problem (2)

Juno lighting makes special lights that are popular in new homes. Juno expects demand for two popular lights to be the following over the next eight weeks.

	WEEK							
	1	2	3	4	5	6	7	8
VH1-234	34	37	41	45	48	48	48	48
VH2-100	104	134	144	155	134	140	141	145

A Key component in both lights is a socket that the bulb is screwed into the base fixture. Each light has one of these sockets. Given the following information, plan the production of lights and purchases of sockets.

	VH1-234	VH2-100	LIGHT SOCKET
On Hand	85	358	425
Q	200(the production lot size)	400 (the production lot size)	500(purchase quantity)
Lead time	1 week	1 week	3 weeks
Safety stock	0 units	0 units	20 units

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ITEM		WEEK 1	2	3	4	5	6	7	8
VH1-234	Gross requirements								
Q=200	Scheduled receipts								
LT=1	Projected available balance								
OH=85	Net requirements								
SS=0	Planned-order receipts								
	Planned-order releases								
VH2-100	Gross requirements								
Q=400	Scheduled receipts								
LT=1	Projected available balance								
OH=358	Net requirements								
SS=0	Planned-order receipts								
	Planned-order releases								
SOCKET	Gross requirements								
Q=500	Scheduled receipts								
LT=3	Projected available balance								
OH=425	Net requirements								
SS=20	Planned-order receipts								
	Planned-order releases								

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LOT Sizing in MRP
MRP OUTPUTS

MRP systems have the ability to provide management with a fairly broad range of outputs. These are often classified as primary reports, which are the main reports, and secondary reports, which are optional outputs.

- **Primary Reports-** Production and inventory planning and control are part of primary reports. These reports normally include the following:
- **Planned orders, Order releases, Changes to planned orders.**
- **Secondary Reports- Performance control, planning, and exceptions belong to secondary reports.**

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Lot Sizing

Determining a lot size to order or to produce is an important issue in inventory management for both independent- and dependent-demand items.
For independent- demand items, managers often use economic order sizes and economic production quantities.
For dependent-demand systems, however, a much wider variety of plans is used to determine lot sizes, mainly because no single plan has a clear advantage over the others.

- The methods used to handle lot sizing range from the complex, which attempt to include all relevant costs, to the very simple, which are easy to use and understand.
- In certain cases, the simple models seem to approach cost minimization although generalizations are difficult.

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Models of Lot Sizing

1) Lot-for-Lot Ordering-

- The order or run size for each period is set equal to demand for that period.
- It minimizes investment in inventory.
- If setup costs can be significantly reduced, this method may approximate a minimum cost lot size.

- **Its two chief drawbacks are –**

1) It usually involves many different order sizes and thus cannot take advantage of the economies of fixed order size.
2) It requires a new setup for each production run.

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Example Problem(1) Lot-for-Lot run Size for an MRP Schedule

(1) Week	(2) Net Requirements	(3) Production Quantity	(4) Ending Inventory	(5) Holding Cost	(6) Setup Cost	(7) Total Cost
1	50	50	0	$0.00	$47.00	$47.00
2	60	60	0	$0.00	$47.00	$94.00
3	70	70	0	$0.00	$47.00	$141.00
4	60	60	0	$0.00	$47.00	$188.00
5	95	95	0	$0.00	$47.00	$235.00
6	75	75	0	$0.00	$47.00	$282.00
7	60	60	0	$0.00	$47.00	$329.00
8	55	55	0	$0.00	$47.00	$376.00

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2) Economic Order Quantity Model-

- The case for lower-level items that are common to different parents and for raw materials.
- However, the more lumpy demand is, the less appropriate such an approach is, because the mismatch in supply and demand results in leftover inventories.

EOQ ------- EPQ
Order Cost ------------ Setup cost

Annual demand based on the 8 weeks = D= (525*52)/8 = 3412.5 Units

Annual holding cost= H= 0.5% * $ 10 * 52 weeks = $2.60 per unit

Setup cost= S= $47 (given)

EOQ = $\sqrt{2DS/H}$ **=** $\sqrt{2*(3412.5)*(\$47)/\$2.60}$ **= 351 units**

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Example Problem(2)

EOQ Run Size for an MRP Schedule

(1) Week	(2) Net Requirements	(3) Production Quantity	(4) Ending Inventory	(5) Holding Cost	(6) Setup Cost	(7) Total Cost
1	50	351	301	$15.05	$47.00	$62.05
2	60	0	241	$12.05	$0.00	$74.10
3	70	0	171	$8.55	$0.00	$82.65
4	60	0	111	$5.55	$0.00	$88.20
5	95	0	16	$0.80	$0.00	$89.00
6	75	351	292	$14.60	$47.00	$150.60
7	60	0	232	$11.60	$0.00	$162.20
8	55	0	177	$8.85	$0.00	$171.05

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3) Least total cost-

- It is a dynamic lot-sizing technique that calculates the order quantity by comparing the carrying cost and the setup(or ordering) costs for various lot sizes and then selects the lot in which are most nearly equal.

4) Least unit cost-

- It is a dynamic lot-sizing technique that adds ordering and inventory carrying cost for each trial lot size and divides by the number of units in each lot size, picking the lot size with the lowest unit cost.

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Example Problem(3)

Least total cost Run Size for an MRP Schedule

WEEKS	QUANTITY ORDERED	CARRYING COST	ORDER COST	TOTAL COST
1	50	$0.00	$47.00	$47.00
1-2	110	$3.00	$47.00	$50.00
1-3	180	$10.00	$47.00	$57.00
1-4	240	$19.00	$47.00	$66.00 1ST ORDER
1-5	335	$38.00	$47.00	$85.00 Least total cost
1-6	410	$56.75	$47.00	$103.75
1-7	470	$74.75	$47.00	$121.75
1-8	525	$94.00	$47.00	$141.00
1-6	75	$0.00	$47.00	$47.00
1-7	135	$3.00	$47.00	$50.00 2nd order
1-8	190	$8.50	$47.00	$55.50 Least total cost

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Example Problem(3)

(1) Week	(2) Net Requirements	(3) Production Quantity	(4) Ending Inventory	(5) Holding Cost	(6) Setup Cost	(7) Total Cost
1	50	335 ✓	285	$14.25	$47.00	$61.25
2	60	0	225	$11.25	$0.00	$72.50
3	70	0	155	$7.75	$0.00	$80.25
4	60	0	95	$4.75	$0.00	$85.00
5	95	0	0	$0.00	$0.00	$85.00
6	75	190	115	$5.75	$47.00	$137.75
7	60	0	55	$2.75	$0.00	$140.50
8	55	0	0	$0.00	$0.00	$140.50

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Example Problem(4)

Least unit cost Run Size for an MRP Schedule

WEEKS	QUANTITY ORDERED	CARRYING COST	ORDER COST	TOTAL COST	UNIT COST
1	50	$0.00	$47.00	$47.00	$0.9400
1-2	110	$3.00	$47.00	$50.00	$0.4545
1-3	180	$10.00	$47.00	$57.00	$0.3167
1-4	240	$19.00	$47.00	$66.00	$0.2750
1-5	335	$38.00	$47.00	$85.00	$0.2537 1ST ORDER
1-6	410	$56.75	$47.00	$103.75	$0.2530 Least unit cost
1-7	470	$74.75	$47.00	$121.75	$0.2590
1-8	525	$94.00	$47.00	$141.00	$0.2686
?	60	$0.00	$47.00	$50.00	$0.7833 2nd order
7-8	115	$2.75	$47.00	$55.50	$0.4326 Least unit cost

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Example Problem(4)

(1) Week	(2) Net Requirements	(3) Production Quantity	(4) Ending Inventory	(5) Holding Cost	(6) Setup Cost	(7) Total Cost
1	50	410	360	$18.00	$47.00	$65.00
2	60	0	300	$15.00	$0.00	$80.00
3	70	0	230	$11.50	$0.00	$91.50
4	60	0	170	$8.50	$0.00	$100.00
5	95	0	75	$3.75	$0.00	$103.75
6	75	115	0	$0	$0.00	$103.75
7	60	0	55	$2.75	$47.00	$153.50
8	55	0	0	$0	$0.00	$153.50

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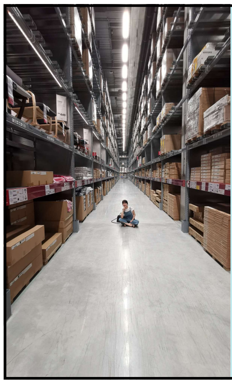

Inventory records file

It is an input as well as an output to the MRP

The inventory records file is used to track information on the status of each item by time period. This includes gross requirements, scheduled receipts, and the expected amount on hand. It includes other details for each item as well, like the supplier, the lead-time, and the lot size

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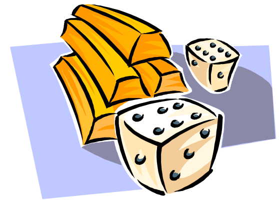
MRP Outputs

- Manufacturin g Orders
- Purchasing Orders
- Various Reports



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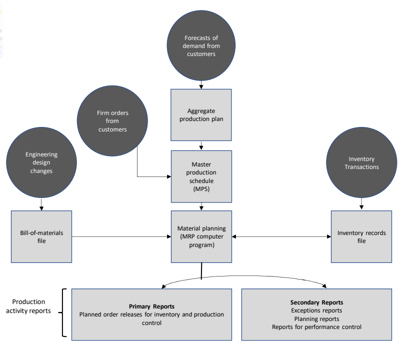

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Summary of MRP

MRP is…

- Computerized Inventory Control
- Production Planning System…
- Schedules Component Items as Needed which will…
- Track Inventory and…
- Help in many other aspects of business…

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